

Modeling miRNA Pressure on Cancer Hallmarks — NotebookLM Slides Companion

HOW TO USE THIS FILE

This file has two parts.

- **PART A — Facts & worked example.** Upload this (or the whole file) to NotebookLM **as an additional source**, alongside MODELING_FRAMEWORK_EXTERNAL.md. It carries the exact counts, the single worked edge, the protein survivors, the trajectory examples, and the 17 orphan edges — so NotebookLM has real numbers to ground on instead of paraphrasing them away.
- **PART B — Per-slide build spec.** Paste this into the NotebookLM **slides “Customize” box** (the instruction field). It is the slide-by-slide ordering with the four insertions, the slide-7 fix, the slide-12 split, and the slide-13 sharpening.

Everything below is grounded in the repo (TCGA-BRCA spine; Normal-like set aside → **1,065 retained tumours**, realized tests on **~1,041**). Numbers re-derive each rebuild; treat ρ/q to 2 sig-figs.

PART A — FACTS & WORKED EXAMPLE (upload as a source)

A0. The universe in numbers (say these out loud)

Quantity	Value
Cohort	TCGA-BRCA primary tumours; 1,065 retained (PAM50 Normal-like 36 set aside); realized coupling on ~1,041 with complete miRNA+mRNA+covariates
miRNA arms (expressible)	721
Curated spine edges	5,108 (miRTarBase functional core)
Target genes under pressure	1,410 (of 4,384 genes in the union of the 50 programs)
Programs	50 MSigDB Hallmark gene sets
PAM50 strata (retained n)	Luminal A 507 · Luminal B 264 · HER2-enriched 93 · Basal-like 197
Healthy reference	GTEx breast 346 (+ DIANA-miTED atlas for coverage repair)
Normal-adjacent (NAT)	104 (same platform; field-effect intermediate state)
Protein cohorts	CPTAC same-patient n=105 + independent n≈101
Independent replication	Buffa 2011, 210 paired tumours, no TCGA overlap
Priors	232 curated drivers (114 oncogene / 106 TSG / 12 dual) · DepMap 96 breast lines · 1,639 TFs

The five resolutions: Universe > Program > Target-set > Gene > Edge.

A1. The master formula (the spine of the whole deck)

$$c(m, g, s) = \text{expr}(m, s) \cdot w(m, g) \quad \text{pressure}(g, s) = \sum_{m \in R(g)} c(m, g, s)$$

- $c(m, g, s)$ = the **edge-resolution pressure atom** (one miRNA m , one gene g , one sample s).
- Two high-level components: $w(m, g)$ = **interaction weight** (static, evidence-based) and $\text{expr}(m, s)$ = **expression component** (dynamic, sample-varying).

- A gene's pressure is the **sum** of c over its regulators $R(g)$.

Component previews (one line each, before the detail slides): - $w(m, g) = \ell(\text{ev_eff}(m, g)) / D(m)$ — evidence numerator ÷ promiscuity denominator, $\ell(n) = \ln(1+n)$. - $\text{expr}(m, s) = \text{sm} \cdot z \cdot \max(x, 0)$ (spine mode) — **share** × **dynamics** × **abundance level**.

A2. THE WORKED EDGE — hsa-miR-17-5p → PTEN (use this ONE edge for slides 7-9 and again at protein)

This single edge threads the whole deck: it is a curated spine edge, it is mRNA-coupled, it is **protein-validated**, and its target **PTEN** is a 91-regulator hub for the gene-aggregation slide. Use it instead of a generic “collagen axis.”

Identity - Edge: hsa-miR-17-5p → PTEN (PTEN = PI3K/AKT tumour-suppressor phosphatase). - Seed family: miR-17-5p / 20-5p / 93-5p / 106-5p / 519-3p (the **miR-17~92 oncomiR cluster**). - Hallmark sets the edge sits in: PI3K_AKT_MTOR_SIGNALING, APICAL_JUNCTION, UV_RESPONSE_DN. - Biology: a proliferation-cluster oncomiR repressing a canonical tumour suppressor — the textbook **proliferation-cluster** → **tumour-suppressor** hub.

(1) Interaction weight $w(m, g)$ — static evidence - Curated assay counts for this edge: **reporter assays $n=1$, protein assays $n=1$** ; passes the high-evidence gate (high_evidence = True). - Evidence numerator (with $\ell(n) = \ln(1+n)$, $\ell(1) = 0.693$):

$$\text{ev} = 3.0 \ell(n_{\text{rep}}) + 2.5 \ell(n_{\text{prot}}) + 1.5 \ell(n_{\text{ma}}) + 1.5 \ell(n_{\text{pert}}) + 0.5 \ell(n_{\text{bind}})$$

Here reporter=1, protein=1 ⇒ leading terms = $(3.0+2.5) \cdot 0.693 \approx 3.81$ (other classes 0 for this edge). - $\text{ev_eff} = \text{ev} + 0.5 \cdot \text{enc}$ (ENCORI crosslink/degradome depth boost where present). - $w(m, g) = \ell(\text{ev_eff}) / \max(D(m), \varepsilon)$ — $D(m)$ is miR-17-5p's per-arm **promiscuity scalar** (total credible evidence mass over all its targets), so a hub miRNA can't dominate any one gene just by being promiscuous. - (Pipeline integer evidence_score for this edge = **9**; the miR-17~92 sibling miR-20a-5p → PTEN = **15**, two studies.)

(2) Expression component $\text{expr}(m, s)$ — dynamic, three subcomponents (spine mode)

$$\text{expr}(m, s) = \underbrace{\text{sm}_{m,s}}_{\text{share among PTEN's regulators}} \times \underbrace{z_{m,s}}_{\text{within-tumour dynamics}} \times \underbrace{\max(x_{m,s}, 0)}_{\text{abundance level}}$$

with $x = \log_2(\text{RPM}+1)$, $z = (x - \mu) / \sigma$ (standardized within scope), and sm = softmax share of miR-17-5p's abundance among the regulators converging on PTEN. (Attribution mode drops z ; acquired mode replaces z by deviation-from-healthy $\text{dev} = x - h$.)

(3) Contribution $c = \text{expr} \cdot w$ — the pressure this edge places on PTEN in sample s .

(4) Realized coupling — does the pressure show up in tumours? Partial Spearman across ~1,041 tumours, miR-17-5p abundance vs **PTEN mRNA**, confounders (purity, HRD/instability, PTEN copy number, proliferation, composition, batch) removed:

Edge	partial- ρ	significance	reading
miR-17-5p → PTEN	-0.29	q_by $\approx 1\text{e-19}$, perm $p \approx 5\text{e-4}$	strong realized repression
miR-20a-5p → PTEN (sibling)	-0.32	FDR-sig	cluster coheres
miR-106b-5p → PTEN (sibling)	-0.34	FDR-sig	cluster coheres
seed-family aggregate → PTEN	—	q_seedfamily $\approx 1.5\text{e-26}$	family-level call
miR-21-5p → PTEN (contrast)	-0.03	n.s. ($q \approx 0.30$)	the <i>famous</i> pair is pressure-present but realization-weak — the platform's point

(5) Protein layer (CPTAC, independent cohort) — PTEN protein residualized on its own mRNA, widest covariate stack: **resid- ρ = -0.34, q = 0.008 (85 candidate cluster arms; strongest in Basal). The proliferation-cluster→TS hub is recovered **at the protein level**.**

Teaching contrast to make explicit: **miR-21**→**PTEN** is the textbook pair but does NOT realize ($\rho=-0.03$); the miR-17~92 cluster does, at mRNA and protein. Realized $\neq$ famous.

A3. PTEN AS A GENE — aggregation in practice (for the new gene slide before slide 10)

PTEN carries **91 regulators**. Aggregation answers two *different* questions, via two modes:

Mode	Formula piece	"Who lands hardest on PTEN?"	Leader
Attributive (no-z: sm·level)	steady-state share	which arm carries the most standing pressure	miR-9-5p , gene-side share 0.080 (#1; rose 0.048→0.080, grip tightening)
Dynamic (spine: sm·z·level)	share $\times$ dynamics	which arm's variation tracks PTEN down	miR-17~92 cluster (17-5p/20a/106b, the coupled arms above)

- The attribution leader (**miR-9-5p**) and the coupling leaders (**miR-17~92**) are *different arms* — exactly why both modes exist.
- Arm-side vs gene-side: PTEN is miR-9-5p's **#1** gene (gene-side share 0.080) but PTEN is a **trivial 0.006** slice of miR-9-5p's own portfolio (arm-side edge_share) — opposite questions, both kept.
- PTEN is a **heavily-contested** gene: it always keeps *some* coupled regulator while individual arms decouple, so no single arm "controls" it. Cohort context: **282 / 1,424** genes are net-repressed in tumour (gene-level partial- ρ of own expression vs aggregate incoming pressure).

A4. PROTEIN REALIZATION — CPTAC (slide 12a, split out)

Three tests on mass-spec proteomes (standardized mRNA & protein):

Test	Quantity	Expected
direct protein	$\rho(\text{pressure, protein})$	negative
mRNA→protein gap	$\rho(\text{pressure, mRNA} - \text{protein})$	positive
protein beyond mRNA	$\rho(\text{pressure, protein} \mid \text{mRNA})$	negative — the key test

Named FDR-significant protein-residual survivors (prospective CPTAC, resid on own mRNA, widest covariates):

Gene ← driver arm	resid- ρ (q)	note
KRT8 ← miR-186-5p	−0.44 (6e-4)	keratin 8
DDAH1 ← miR-210-3p	−0.42 (9e-4)	hypoxia oncomiR
BMP1 ← miR-194-5p	−0.40 (1e-3)	also a miR-29 target
PEBP1 ← miR-155-5p	−0.39 (1e-3)	RKIP, MAPK brake
ARHGDI ← miR-151a-5p	−0.39 (1e-3)	Rho-GDI
MINPP1 ← miR-30b-5p	−0.35 (6e-3)	inositol phosphatase
PTEN ← miR-17-5p	−0.34 (8e-3)	the cluster→TS hub (our worked edge)
HMG2 ← miR-23a-3p	−0.33 (1e-2)	chromatin
ZYX ← miR-16-5p	−0.31 (2e-2)	zyxin

Headlines: dozens of genes FDR-significant; repression is **predominantly mRNA-mediated** (residualizing on mRNA collapses most of it, leaving a small beyond-mRNA residual); survivors are **known, subtype-structured** edges incl. the PTEN hub; a **decoy control** confirms target-specificity (each arm hits its cognate protein, not the proteins it does not target); survives batch adjustment. **Caveat to state:** prognostically the EMT signal runs through **LOX / ECM, not ZEB1/2** (per the outcome deep-dive) — do **not** call ZEB1/2 the protein driver.

A5. HEALTHY → NAT → TUMOUR TRAJECTORY (slide 12b, split out)

Three states: **GTE** healthy (346) → NAT (104) → **tumour**. Two acquired axes (union call): - **rank delta** $d_{HT} = r_{tumour} - r_{healthy}$ (saturating, cross-platform-robust), - **magnitude** $\log_2 fc = \hat{x}_{tumour} - b_{healthy}$ (recovers the surge). - BC oncomiR surge example: **miR-183 / 182 / 375 / 93 / 141, +9 to +13 log2**.

Released brakes — repression present in healthy, gone in tumour (R00 pattern): **189 edges / 109 arms**.

Edge	ρ healthy → tumour	biology
miR-193a-5p → ERBB2	-0.28 → +0.07	HER2 brake released
miR-224-5p → CXCR4	-0.25 → +0.02	metastatic-homing brake
miR-22-3p → MMP14 · miR-708-5p → MMP2	-0.30 → +0.19 · -0.31 → +0.03	invasion (MMP de-repression)
miR-326 / miR-23b-3p → NOTCH2	-0.33 / -0.30 → ~0	Notch de-repression
miR-363-3p → BCL2L1(BIM)	-0.28 → ~0	apoptosis brake

Theme: healthy breast actively holds oncogenic/invasion targets (ERBB2, CXCR4, MMP2/14, NOTCH2) under miRNA repression; the tumour **releases the brake**.

Acquired-and-realized — pressure rises *and* coupling turns repressive: **640 edges / 129 arms**.

Edge	ρ healthy → tumour	biology
miR-744-3p → HLA-G + → TGFB1	+0.01 → -0.10 · +0.01 → -0.24	novel immune axis (MHC tolerance ligand + TGFβ)
miR-590-5p → RBPJ	+0.02 → -0.11	acquired Notch-effector repression
miR-301a-3p → BTG1	-0.11 → -0.20	strengthening anti-proliferative-TS repression

(Field-established 0RR: miR-21-5p→TGFB3, miR-375→AHR, miR-25→PTEN.)

A6. ORPHAN DISCOVERY SURFACE — and THE 17 (slide 13, sharpened)

Orphan edge = TargetScan/ENCORI-predicted, **absent from miRTarBase functional curation** — deliberately kept out of the spine so it is an unbiased screen.

The funnel:

20,663 orphan edges screened (TargetScan ∪ ENCORI, no functional MTI)

→ 15,512 testable vs CPTAC protein

→ 539 protein anti-correlation candidates

→ 182 translational (beyond-mRNA residual)

→ 93 translational AND TCGA-replicated (76/93 fully uncured)

- **Triple-cohort (TCGA mRNA · CPTAC protein · Buffa mRNA):** of **492 genuine orphans**, **127** were CPTAC-sig + TCGA-replicated + Buffa-testable; **108 (85%)** keep a negative sign in Buffa (47 reach FDR) — far above the 50% chance rate; led by **miR-29→ECM (34/108)**.
- **De-novo anchor:** the **miR-29 → collagen/ECM** axis re-emerges *at uncured targets* (miR-29a→COL11A1 ts_weight 0.80, 15 CLIP experiments; →COL6A2, →BMP1) — credibility check.

The 17 composition-robust core (MH-39). Re-test the prediction-only tier (TargetScan sequence-only — **no CLIP, no curation**) after adding PAM50 + ESTIMATE immune/stromal + endothelial + target-copy-number covariates: **17 of 24 survivors hold** (negative beyond-mRNA protein_resid at $q < 0.1$ under the widest covariate set; **11/17 at $q < 0.05$**). The 7 drops are exactly the composition artifacts (VCL×2, TPM4, and the hematopoietic miR-142 cluster). **13 target genes, 17 edges:**

Target (function)	TargetScan-only arm(s)	protein ρ	resid- ρ (q)
PMM1 — mannose metabolism	miR-195-5p · miR-16-5p · miR-15a-5p	-0.40 · -0.39 · -0.39	-0.40 (9e-4) · -0.31 (.012) · -0.30 (.012)
PPM1A — PP2C phosphatase (TGF-β/AKT brake)	miR-19b-3p · miR-19a-3p	-0.30 · -0.32	-0.28 (.016) · -0.23 (.042)

Target (function)	TargetScan-only arm(s)	protein ρ	resid- ρ (q)
PURA — transcription / replication	miR-532-5p · miR-452-5p	-0.45 · -0.33	-0.19 (.093) · -0.18 (.093)
KIF1B — kinesin, pro-apoptotic	miR-15a-5p	-0.37	-0.30 (.012)
NCK2 — signalling adaptor	miR-142-3p	-0.07	-0.31 (.012)
VCAN — versican (ECM)	miR-107	-0.52	-0.27 (.018)
MAPRE1 — EB1, microtubule +end	miR-429	-0.25	-0.27 (.018)
NUDT4 — Nudix hydrolase	miR-183-5p	-0.16	-0.26 (.023)
RAB1A — ER→Golgi traffic	miR-101-3p	-0.33	-0.24 (.041)
CERCAM — endothelial adhesion	let-7g-5p	-0.35	-0.22 (.057)
SKP1 — SCF ubiquitin-ligase core	miR-452-5p	-0.38	-0.21 (.066)
DR1 — NC2 transcriptional repressor	miR-182-5p	-0.34	-0.19 (.093)
GLUD1 — glutamate dehydrogenase	miR-30e-5p	-0.28	-0.18 (.093)

What to emphasize about the 17: - **Counts:** 17 edges, 13 genes, **11/17 at $q < 0.05$** ; all from sequence-only predictions that survived a full composition + copy-number confounder stack. - **Two seed-family convergences lead and are CAF-neutral / method-independent:** the triple-hit **miR-15/16/195 → PMM1** (a **metabolic convergence**, mRNA-flat but protein-strong = translational) and **miR-19a/b → PPM1A**. - **NCK2 is the instructive single case:** direct protein $\rho \approx 0$, yet the **beyond-mRNA residual is strong (-0.31, $q = .012$)** — a translational-layer signal bulk mRNA cannot see. - **General character of the genes:** mostly **metabolism** (PMM1, GLUD1, NUDT4), **signalling brakes** (PPM1A, NCK2, SKP1, RAB1A), **transcription/replication** (PURA, DR1, MAPRE1), plus single ECM (VCAN), apoptosis (KIF1B), endothelial (CERCAM) — i.e. NOT the curated collagen crowd; these are *new* metabolic/signalling nominations. - **Honest status:** these are **wet-lab nominations** (a prioritized CLIP/luciferase queue), not validated edges — protein anti-correlation $\neq$ binding, co-expression confound persists, and seed-family arms (e.g. miR-15/16) cannot be separated.

PART B — PER-SLIDE BUILD SPEC (paste into the slides “Customize” box)

Build a ~45-minute, ~18-slide work-in-progress lab talk from the two sources. Be quantitative: when a number, formula, or gene appears in the sources, **put it on the slide** — do not round to “several” or paraphrase it away. Use the companion file’s worked edge (**miR-17-5p → PTEN**) as a through-line. Produce exactly these slides in this order:

1. **Title** — “Modeling miRNA Repressive Pressure on Cancer Hallmark Programs (breast cancer).”
2. **The three rungs** — pressure (potential) vs coupling (realized) vs causation; why they are kept distinct.
3. **Data & cohort, in numbers** — 1,065 retained tumours (36 Normal-like set aside), ~1,041 for coupling; 721 arms, 5,108 edges, 1,410 genes, 50 programs; PAM50 507/264/93/197; GTEx 346, NAT 104, CPTAC 105 + 101, Buffa 210.
4. **Five resolutions + the interaction universe** — Universe > Program > Target-set > Gene > Edge; spine = miRTarBase functional core; orphans held out.
5. ★ **NEW — The master formula at a glance.** Show $c(m,g,s) = \text{expr}(m,s) \cdot w(m,g)$ and $\text{pressure}(g,s) = \sum c$. Name the **two high-level components**: $w(m,g)$ (static evidence weight) and $\text{expr}(m,s)$ (dynamic expression). Give the one-line previews $w = \ell(\text{ev_eff})/D(m)$ and $\text{expr} = \text{sm} \cdot z \cdot \text{level}$. Say “the next slides open each component.” (Insert this BETWEEN current slide 4 and current slide 5.)
6. **The evidence gate & weighting rationale** — high-evidence gate; evidence numerator (assay-directness weights 3.0/2.5/1.5/1.5/0.5, log-damping, weak-evidence discount) ÷ promiscuity denominator $D(m)$; weights are a transparent prior, not a fit.
7. ★ **FIX — Worked edge weight: miR-17-5p → PTEN (ONE specific edge, not “the collagen axis”).** An edge is one miRNA → one gene, so pick this one. Show its real assay counts

(reporter=1, protein=1, high-evidence), the populated ev numerator (≈ 3.81 from the two direct-assay terms), and $w = \ell(\text{ev_eff})/D(\text{miR-17-5p})$. Name the seed family (miR-17~92) and the biology (oncomiR cluster $\rightarrow$ tumour-suppressor PTEN).

8. ★ **FIX — The expression component, whole then parts.** First show the **entire** spine expression formula $\text{expr}(m,s)=sm \cdot z \cdot \max(x,0)$. THEN break it into its three subcomponents: z (within-tumour dynamics), sm (softmax share among a gene's regulators), $\max(x,0)$ (abundance level). Note the two sibling modes (no- z attribution; deviation-from-healthy acquired).
9. ★ **NEW — Worked edge, expression $\rightarrow$ contribution.** Same miR-17-5p $\rightarrow$ PTEN: walk its three expression subcomponents, then combine with the weight from slide 7 to produce the **contribution $c = \text{expr} \cdot w$** — the pressure atom this edge places on PTEN. (Insert AFTER slide 8.)
10. ★ **NEW — Worked edge, realized coupling.** Same edge: partial Spearman across $\sim 1,041$ tumours of miR-17-5p abundance vs PTEN mRNA, confounders removed $\rightarrow \rho = -0.29$ (**q_{by} $\approx 1e-19$**); siblings miR-20a/106b also negative; seed-family $q \approx 1.5e-26$. **Contrast: miR-21-5p $\rightarrow$ PTEN $\rho = -0.03$, n.s.** — the famous pair does not realize. (Insert AFTER the slide above, BEFORE the gene-aggregation slide.)
11. ★ **NEW — PTEN as a gene: aggregation in practice.** PTEN = **91 regulators**. Show pressure aggregated two ways: **attributive (no- z)** leader **miR-9-5p** (gene-side share 0.080, #1) vs **dynamic (spine- z)** coupling leaders **miR-17~92**; note arm-side vs gene-side (PTEN is miR-9-5p's #1 gene but a 0.006 slice of its portfolio). (Insert BEFORE the original gene-aggregation formula slide.)
12. **Gene aggregation formula** (original slide 10) — $\text{pressure}(g) = \sum_m c$; share & specificity; arm-side vs gene-side shares; sum (not mean) is the default roll-up.
13. **Architecture** — topology influence (OmniPath signed propagation) + program-polarity prior + gene-role malignancy; the result that miRNAs turn malignant mainly by **brake-release on suppressive programs**.
14. ★ **SPLIT 12a — Protein realization (CPTAC).** Three tests (direct / mRNA-protein gap / **protein beyond mRNA**). Named FDR-significant survivors: **KRT8, DDAH1, BMP1, PEBP1, ARHG-DIA, MINPP1, PTEN (the hub, -0.34 q=.008), HMG2, ZYX**. Headlines: dozens FDR-sig, **predominantly mRNA-mediated**, target-specific (decoy control), batch-robust. **Do not call ZEB1/2 the driver — the EMT prognostic signal is LOX/ECM.**
15. ★ **SPLIT 12b — Healthy $\rightarrow$ NAT $\rightarrow$ tumour trajectory.** Three states (GTEx 346 $\rightarrow$ NAT 104 $\rightarrow$ tumour); two acquired axes (rank delta + log2fc); oncomiR surge miR-183/182/375/93/141 (+9..+13 log2). **Released brakes (189 edges/109 arms): miR-193a $\rightarrow$ ERBB2, miR-224 $\rightarrow$ CXCR4, miR-22/708 $\rightarrow$ MMP14/MMP2, miR-326/23b $\rightarrow$ NOTCH2.** Acquired-realized (640/129): **miR-744-3p $\rightarrow$ HLA-G + TGFB1** (novel immune axis).
16. ★ **SHARPEN 13 — Orphan discovery surface + the 17.** Show the funnel (20,663 $\rightarrow$ 15,512 $\rightarrow$ 539 $\rightarrow$ 182 $\rightarrow$ 93; triple-cohort 108/127). Then show **the 17 composition-robust edges explicitly** (the companion's 13-gene table) with anti-correlation values and gene annotations. Call out: **11/17 at q<0.05**; the metabolic lead **miR-15/16/195 $\rightarrow$ PMM1; NCK2** as translational-only (direct $\rho \approx 0$, residual -0.31); general character = metabolism + signalling brakes, not the curated ECM crowd; miR-29 $\rightarrow$ collagen as the de-novo anchor. Label all 17 **wet-lab nominations**.
17. **Validation ladder & honest limits** — recovered (textbook edges unprompted) $\rightarrow$ mRNA coupling $\rightarrow$ state-resolved trajectory $\rightarrow$ protein layer $\rightarrow$ independent cohort $\rightarrow$ wet-lab queue; limits: bulk-tissue ceiling, metabolite-mediated regulation invisible to topology, seed-family non-identifiability, pressure construction non-unique.
18. **Where the lab comes in** — 2-3 specific open questions (e.g. which of the 17 to CLIP/luciferase first; a matched-NAT-miRNA + proteome cohort to power acquired $\rightarrow$ protein; resolving seed-family attribution).

Style: define every symbol in plain words the first time. Flag preliminary results as preliminary. End slide 18 by asking the lab the open questions.